

Lecture 3: Lorentz Transformations

Based on Morin's *Introduction to Classical Mechanics*, Ch. 11.4–11.5

Outline

1. Recap lecture 2
2. Diagram toolkit for transformations
3. Lorentz Transformations
4. Einstein Velocity Transformations

1 Recap Lecture 2

1.1 Events

An **event** is a specific point in spacetime, defined by:

$$(t, x, y, z).$$

It occurs at a particular location and instant in time (which we have to add to our 'position' vector). Events are the basic entities in relativity. All observers agree on whether two events occurred at the same place and time in their own frame, but may disagree on the simultaneity of events between frames. This is the reason why time has to be added. Since this is time frame dependent (as we saw in lecture 2), we refer to this time as **coordinate time**. Since $\gamma \geq 1$, the clock for an observer at rest runs faster ($t \geq t'$). They are equal only if two observers are at rest with respect to each other (i.e., $S = S'$).

1.2 Time Dilation

Derived from the light clock in lecture 2:

$$t = \gamma t', \quad \text{where } \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \quad (1)$$

Note that the above is assuming there is some zero point time, since this relation really only applies to intervals, i.e. $\Delta t \equiv t_2 - t_1$, where $t_2 > t_1$. Hence, the above implicitly assumes $t_1 = 0$. I am writing it like this here, since Morin tends to drop the Δ .

1.3 Muon Decay

Muons created in the upper atmosphere have a half-life time $t'_{1/2} = 1.5 \mu s$ in their rest frame S' . They travel at $v = 0.99c$. These cosmic muons are created at a height of 60 km in the atmosphere. In a Galilean Universe, we would argue that it would take

$$\Delta t = \Delta h/v = 60 \text{ km}/(0.99c) \sim 200 \mu s \quad (2)$$

to reach Earth. Given the half-life, you would thus get that

$$N \times \left(\frac{1}{2}\right)^{200/1.5} = 1 \rightarrow N = 2^{200/1.5} \sim 10^{40}. \quad (3)$$

So 1 in 10^{40} muons is expected to reach Earth.

In SR, the lifetime is measured in the frame that is at rest with respect to the Muon S' . So we need to fold in time dilation (or length contraction, both work).

In the Earth's frame S , the lifetime of the muon is actually longer by

$$\gamma(v = 0.99c) \simeq 7 \rightarrow t_{1/2} \simeq 10.5 \mu s. \quad (4)$$

If SR is correct, we thus conclude that

$$N \sim 2^{200/10.5} \sim 5 \times 10^5. \quad (5)$$

So over 1 in 500.000 reaches Earth, which matches observations.

1.4 Length Contraction

Proper length L_0 : Length measured in the frame where the object is at rest.

We derived that the length of an object that is moving with respect to an observer at rest is contracted:

$$L = L_0 \sqrt{1 - v^2/c^2} = \frac{L_0}{\gamma}. \quad (6)$$

Since $\gamma \geq 1$, $0 \leq \gamma^{-1} \leq 1$, i.e., the length is contracted. Only for a (hypothetical) object that would move at c , the length would be contracted to zero.

Example 2: High-Speed Train

A train of proper length $L_0 = 500$ m moves at $v = 0.8c$. Find its length in the ground frame.

$$\gamma = \frac{1}{\sqrt{1 - 0.64}} = \frac{1}{0.6} \approx 1.667$$
$$L = \frac{500}{1.667} \approx 300 \text{ m}$$

Note that while the train is already moving close to the speed of light, length contraction is still relatively minor. Since the Lorentz factor scales as the square root of the velocity, the velocity has to get exponentially close to c to get large effects. It is also the reason why, in everyday life, we have no experience with relativistic effects.

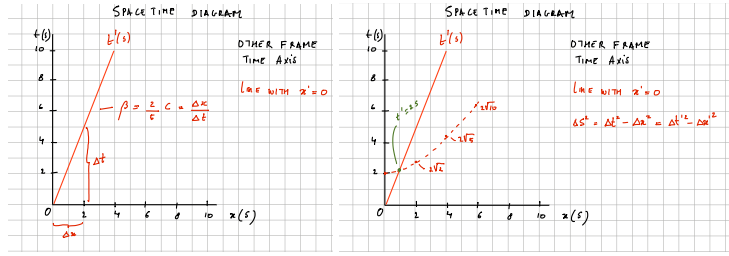


Figure 1: A two-observer diagram. The unprimed frame S uses the vertical t axis and horizontal x axis. The moving frame S' uses tilted t' and x' axes.

2 Diagram toolkit for transformations

Before deriving the Lorentz transformations algebraically, it is useful to know what the answer should look like on a spacetime diagram. We use units with $c = 1$, so light rays always have slope ± 1 .

- The t' axis is the worldline of the origin of S' . If S' moves with velocity v in S , then this line satisfies $x = vt$.
- The x' axis is the set of events that are simultaneous with $t' = 0$ in the moving frame. It tilts by the same amount toward the light cone as the t' axis.
- The light cone is not tilted by a change of inertial frame. This is the geometric statement that all inertial observers measure the same speed of light.
- Lines of constant x' are parallel to the t' axis; lines of constant t' are parallel to the x' axis.

The important lesson is that a Lorentz transformation is not a rotation in the Euclidean sense. It tilts the time and space axes toward the light cone while preserving the spacetime interval. This picture should be kept next to the algebra below: the equations tell us how to compute coordinates, while the diagram tells us what the computation means.

3 Lorentz transformations

In Moore, the Lorentz transformations are derived only in **Chapter 5**. In Morin, they are derived in **section 5** of the first chapter. We will first use the spacetime diagram above as a guide, and then follow Morin's algebraic derivation. We will use some of the things we have just learned about clocks and lengths.

A key point in SR is that, unlike Galilean Relativity, space and time are interwoven. We know a priori that we move both through space and time. While time is absolute and unchanging for observers in different inertial frames, this will not hold in SR. From Galilean transformation rules, we know that a transformation is simply some (mathematical) relation between the coordinates in one frame S and another frame S' . Suppose that S' is moving to the right (positive velocity), which we will call x . We will use (un)primed coordinates (t', x') ((t, x)) for frame S' (S). We will ignore the two other spatial

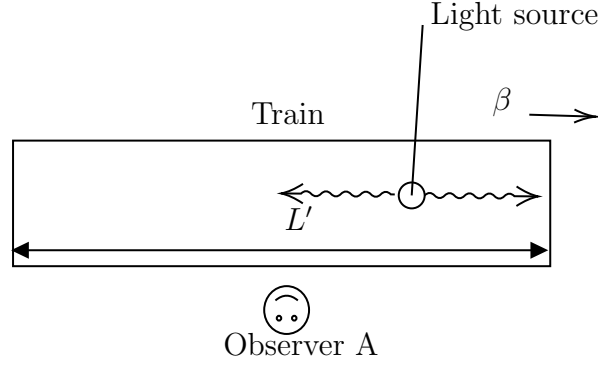


Figure 2: A train of length L' (as measured at rest in frame S') is moving to the right with velocity v . A source on the train is emitting two pulses. They arrive at the rear and front of the train at the same time, as observed (by observer 'A') in the rest frame S . Figure taken from R2, 2023-2024.

directions, as we have learned that space (length) is not affected in those directions if there is no motion in that direction. Given the example of Galilean transformations, we are going to make an ansatz about the transformation rules in SR, which are:

$$\Delta x = A\Delta x' + B\Delta t' \quad (7)$$

$$\Delta t = C\Delta t' + D\Delta x'. \quad (8)$$

Our goal will be to determine the constants A, B, C , and D . Like Galilean transformations, these functions will depend on $v = v_x$, the velocity of frame S' as measured in frame S .

To determine these constants, we can take limiting cases in which we know the answer. First, consider no displacement in frame S' . In this case, we can use our knowledge of time dilation:

$$\Delta t = \gamma\Delta t'. \quad (9)$$

In other words, $C = \gamma$ (that was easy!).

Next, let us consider length contraction. In that case, we should set $\Delta t' = 0$, while

$$\Delta x' = \gamma^{-1}\Delta x. \quad (10)$$

Hence $A = \gamma$.

Now things will get a little bit trickier. Suppose I am an observer in frame S and I am not moving ($\Delta x = 0$). What is my velocity in the other frame? That is easy, it should be $-v$ (since I am at rest and the other frame S' has velocity v). The velocity in that other frame should be determined via $\Delta x'/\Delta t' = -v$. This implies that $B/A = v$ or $B = \gamma v$.

Finally, we need to derive a relation that is valid when $\Delta t = 0$. Morin refers to this effect as rear-clock ahead, which states that a clock (on a train) that is at the back (rear) will be ahead of a clock at the front, when we observe something that is simultaneous in a frame at rest with respect to this train. I alluded to this example in lecture 2. It goes as follows.

Consider the setup as shown in Fig. 3, where a train is moving with velocity v . The train has length L' (as measured at rest – proper length). Two light rays/pulses are

submitted to the rear and front of the train, which arrive simultaneously for observer A ($\Delta t = 0$). Light rays appear to move $c + v$ to the back of the train and $c - v$ to the front of the train. This is ok, since it is really the distance between the location of the emitted pulse and the front and rear of the train that is changing; the rear is moving towards this point with velocity v , while the front is moving away from this point with velocity v .

Now that we have convinced ourselves that this makes sense (Morin comments on this also), we can also write a relative relation between the distances to the front and rear of the train from the emitter:

$$\frac{D'_{\text{back}}}{D'_{\text{front}}} = \frac{c + v}{c - v} = \frac{1 + \beta}{1 - \beta}, \quad (11)$$

where I have divided by c and defined $\beta = v/c$ for now. At the same time, we know:

$$D'_{\text{back}} + D'_{\text{front}} = L'. \quad (12)$$

Hence

$$D'_{\text{back}} = \frac{1 + \beta}{1 - \beta} D'_{\text{front}} \quad (13)$$

and

$$D'_{\text{back}} + D'_{\text{front}} = \left(\frac{1 + \beta}{1 - \beta} + 1\right) D'_{\text{front}} = \left(\frac{1 + \beta}{1 - \beta} + \frac{1 - \beta}{1 - \beta}\right) D'_{\text{front}} = \frac{2}{1 - \beta} D'_{\text{front}}. \quad (14)$$

This tells us that $D'_{\text{front}} = L'(1 - \beta)/2$ and

$$D'_{\text{back}} = L' - D'_{\text{front}} = L' - \frac{L'(1 - \beta)}{2} = L'\left(1 - \frac{1 - \beta}{2}\right) = \frac{L'(1 + \beta)}{2}. \quad (15)$$

Then we compute the ‘extra’ distance the light has to move to the back of the train:

$$D'_{\text{back}} - D'_{\text{front}} = L'\frac{1 + \beta}{2} - L'\frac{1 - \beta}{2} = \beta L'. \quad (16)$$

Putting c back in and realizing (rear-clock-ahead):

$$\Delta t' = t_{\text{front}} - t_{\text{back}} = -(D'_{\text{back}} - D'_{\text{front}})/c, \quad (17)$$

we find:

$$\Delta t' = -v\Delta x'/c^2. \quad (18)$$

This allows us to find that $D/C = v/c^2$ and hence $D = \gamma v/c^2$. The Lorentz transformations are thus given by:

$$\Delta x = \gamma(\Delta x' + v\Delta t') \quad (19)$$

$$\Delta t = \gamma(\Delta t' + v\Delta x'/c^2), \quad (20)$$

and (for completeness)

$$\Delta y = \Delta y' \quad (21)$$

$$\Delta z = \Delta z'. \quad (22)$$

These transformation rules are the cornerstone of SR, and from these, everything about SR can be derived (including, e.g., the velocity transformation as we will see soon). In SR units, $c = 1$ and the transformations become completely symmetric, showing indeed that space and time are treated identically in SR. Both spatial and temporal intervals are frame dependent, but we can relate them using the above equations. We will later discuss that in fact we can define something that is frame independent (namely the (spacetime) interval – also proper time), but such a quantity necessarily will have to depend on both spatial and temporal intervals. The existence of such an invariant interval (and its definition) allows us to derive the Lorentz transformation in a different way (something we might discuss at some point, see e.g., Lecture 8 from last year and R2 from 2023-2024).

We can consider the limit $v \rightarrow 0$. In that case, we know $\gamma \rightarrow 1$ and $\Delta t = \Delta t'$ (as $v/c^2 \ll 1$) while $\Delta x = \Delta x' + v\Delta t'$, i.e., the Galilean transformations.

3.1 Matrix notation

We can write the Lorentz transformation in matrix notation if we use SR units:

$$\begin{pmatrix} \Delta x' \\ \Delta t' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta t \end{pmatrix}, \quad (23)$$

with $\beta = v/c$. Note that I have considered the inverse transformations (going from frame S to S'). In that case we pick up a minus sign in the velocity, since from the perspective of an observer at rest in frame S' , anyone comoving¹ with frame S is now moving in the opposite direction. This looks very similar to a rotation matrix. Coordinates in a (Cartesian) plane x, y can be related to coordinates in a plane that is rotated around its origin with an angle α

$$\begin{aligned} x'_A &= x_A \cos \alpha + y_A \sin \alpha, \\ y'_A &= -x_A \sin \alpha + y_A \cos \alpha \end{aligned}$$

This can be written as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (24)$$

We will discuss this analogy later when we talk about rapidity in lecture 5. This can be particularly useful if you want to show that applying n (with $n > 1$) Lorentz transformations can still be expressed as a Lorentz transformation, albeit with the appropriately transformed velocity. There are multiple old exam questions on this topic, which involve pretty tedious algebra (see e.g. R3, 2023-2024, Question 1). With rapidity, such problems become trivial.

3.2 Fundamental effects revisited

3.2.1 Loss of simultaneity

If two events (at e.g., coordinates (t'_1, x'_1) , (t'_2, x'_2)) occur simultaneously in frame S' , we know $(\Delta t', \Delta x') = (0, \Delta x')$, i.e. $t'_1 = t'_2$. Using Eq. (20), we find that in S , $\Delta t = \gamma v \Delta x / c^2$. Unless $\Delta x' = 0$, events would therefore not happen simultaneously in frame S .

¹I will often use comoving, which often is short-hand for stating that an observer is at rest with respect to some coordinate frame, and moving with (=comoving) the frame as observed from any other inertial reference frame.

3.2.2 Time Dilation

Now consider an events that happen on the same location in frame S : ($\Delta t, \Delta x = 0$). In frame S , we can use Eq. (20) to find that $\Delta t = \gamma \Delta t'$. This is by definition time dilation, and according to an observer in frame S , a clock in frame S' runs slow.

As Morin points out, the opposite is also true, i.e., $\Delta t' = \gamma \Delta t$. This appears to be a contradiction. However, this is, in my view, the crux of relativity; who infers the other clock running slow depends on who you assume is moving w.r.t. an observer at rest. For the one that is ‘moving’, by definition, their displacement should be non-zero. So in the first case, we assume they are at rest in their frame (S') (and hence they are moving in the frame comoving with the observer, i.e., frame S). And $\Delta x' = 0$. To obtain the opposite, $\Delta t' = \gamma \Delta t$, you will assume the observer in frame S is at rest and hence moving w.r.t. frame S' , $\Delta x' = 0$. Whether a clock actually runs slow requires us to ‘check’ the clock of the other observer and compare it to our own at least twice: once to set the initial measurement, and once to set the final measurement. To compare clocks after motion, consider the following two possibilities. The first is the trivial possibility. For both observers (let’s call them A and B), $\Delta x = \Delta x'$: they are comoving. In that case, we can compare clocks at all times, assuming that they started at the same position ($x_1 = x'_1$) and $\Delta t = \Delta t'$ (or $\gamma = 1$). Another possibility to compare clocks is when we assume, e.g., $\Delta x = 0$ (observer A in frame S is at rest) and the other observer (B) is moving away (with constant velocity v_B) from A at $t_1 = t'_1 = 0$. To compare clocks and check which clock runs slow, either A or B needs to start moving, because they need to be at the same spatial position ever to be able to compare clocks. So observer B could go back, therefore changing its velocity from e.g. $+v_B$ to $-v_B$, or observer A takes off and moves towards B with $v_A > v_B$ (to catch up with B). This means that either A or B will have to accelerate, and the clock they are carrying will no longer be inertial. The result is that the person who will perform the acceleration will have a clock that runs slow, as we discussed in the twin paradox. There are some other possibilities, such as both moving away or them moving towards each other, but broadly speaking, these two situations carry the key idea of time dilation. Note that in all these examples, one assumes initially that clocks are synchronized at the beginning, and at that instant, one of the observers has a velocity that is constant. In reality, of course, this is not possible. In the twin paradox example, the twin moving away will have to accelerate (from having no velocity on Earth, to moving towards some distant planet). This should already suffice to make their clock non-inertial. But to compare actual times still requires both to be at the same spatial position. So, to truly ‘experience’ time dilation requires at least two measurements.

3.2.3 Length Contraction

Recall that an object’s length in a given inertial frame is defined to be the distance between any two simultaneous events that occur at its ends. Hence, we need to be concerned with measurements of $\Delta t'$ that are zero in one frame. Let us consider an object of length L' in frame S' . To find the length of that object in frame S , we set $\Delta t = 0$ and use inverse of Eq. (19) to find $\Delta x' = \gamma \Delta x$ which implies $L = \gamma^{-1} L'$, the length of an object of length L' in frame S' is contracted in frame S . The reverse is also true (length L in frame S appears constructed in frame S' , something that can be shown using a **two observer** diagram very easily. We will come back to this once we introduce

diagrams.

4 Einstein velocity transformations

We have shown how coordinates transform in the previous section. Here, we aim to show how velocities transform. Take two frames S and S' , and S' is moving with velocity u (in the positive x direction). An object in frame S' is moving with velocity v' in that frame. What is the velocity of that object as observed in S ? We can simply use the LT we just derived. In S the velocity can be written as $v \equiv \frac{dx}{dt} = \frac{\Delta x}{\Delta t}$:

$$v = \frac{\Delta x}{\Delta t} = \frac{\gamma(\Delta x' + u\Delta t')}{\gamma(\Delta t' + u\Delta x'/c^2)} = \frac{\frac{\Delta x'}{\Delta t'} + u}{1 + u\frac{\Delta x'}{\Delta t'}/c^2} \equiv \frac{v' + u}{1 + uv'/c^2}, \quad (25)$$

where I used that $\Delta x'/\Delta t' = v'$. This is the longitudinal velocity addition rule (since the motion of the object is parallel to the motion of the frame S'). Note that if either $v' = c$ or if $u = c$, $v = c$. This should be the case! One of the postulates of SR is that nothing can move faster than the speed of light, even something that is already moving with the speed of light in a frame that is at non-zero speed. We could also interchange v' and u , it does not matter whether you interchange the motion of the frame and the object within that frame. If both u and v' are less than c , $v < c$. In the case we take small velocities compared to c , the denominator $1 + uv'/c^2 \rightarrow 1$ and $v \simeq v' + u$, i.e., the classical result.

Example: In the book, there is an example of two trains (a and b) of length L . Both trains are moving to the right, where $v_a > v_b$ (as measured from a frame at rest) and the train a will overtake train b . The first question is how long it takes for train a to overtake train b . In any such problems, you should always first pick a frame. Morin considers the frame comoving with train b , but one should be able to find the same solution if one considers any other inertial frame. The comoving frame should have velocity $-v_b$ for the train to be at rest. The velocity addition rule then states:

$$v'_a = \frac{v_a - v_b}{1 - v_a v_b / c^2}. \quad (26)$$

In the frame comoving with train b , train a will be contracted. The contraction is given by:

$$L' = \gamma^{-1} L = \sqrt{1 - v_a'^2 / c^2} L. \quad (27)$$

In order to overtake train b , the total distance that train a needs to travel is its own length L' and the length of train b , L , or $D = L + L'$. The time it therefore takes to overtake is given by:

$$\Delta t = \frac{D}{v'_a} = \frac{L(\gamma^{-1} + 1)}{v'_a}. \quad (28)$$

Inserting the example values for $v_a = 4c/5$ and $v_b = 3c/5$, you will see this equals $\Delta t = 5L/c$.

The second question appears more complicated, but what matters again is to choose a clever frame. Consider an observer O inside train b that starts walking from left to right

as soon as the front of train a reaches the rear of train b (which he calls event E_1) and O reaches the front of train b , as soon as the rear of train a passes the front, event E_2 . What is the time between these two events as measured by O ? Morin chooses that frame where the observer O is at rest. In that frame, one train is moving to the right, and the other is moving to the left. However, given the fact that this observer starts to move once the front of train a reaches the rear of train b and reaches the front of train b once the rear of train a passes, implies that in the frame where this observer is at rest, both trains move with the same speed (but in opposite direction), otherwise the above requirement would not be met. To compute the time between events, we need to know this speed, which we call v_O . To find this speed, we can use the fact that two objects with the same but opposite speeds can simply use the Einstein velocity transformation, equating the speed of the frame with that of the object. The frame for which this is true is the frame where O is at rest! At the same time, the relative speed between the two trains was already computed before; it was given by the velocity addition formula. Equating the two

$$\frac{2v_O}{1 + v_O^2/c^2} = \frac{v_a - v_b}{1 - v_a v_b/c^2}. \quad (29)$$

This is a quadratic equation for v_O . The physical root, which satisfies $|v_O| < c$ and has the required direction of motion, is

$$v_O \rightarrow \frac{-\sqrt{v_a^2 v_b^2 - v_a^2 c^2 - v_b^2 c^2 + c^4} - v_a v_b + c^2}{v_a - v_b}. \quad (30)$$

Using $v_a = 4c/5$ and $v_b = 3c/5$, gives $v_O = c/5$. This is associated with a Lorentz factor $\gamma_O = 5/(2\sqrt{6})$. Thus, in the observer's frame, the length of the two trains is contracted, and the total time between E_1 and E_2 as observed by O is given by:

$$\Delta t_O = \frac{\gamma_O^{-1} L}{v_O}, \quad (31)$$

which in this example results in $\Delta t_O = 2\sqrt{6}L/c$.

We discussed earlier that LT only applies to spatial coordinates that are parallel to the direction of motion. We mentioned that this does not apply to velocities, since time is always transformed if two frames have a relative motion, and velocity depends on both space and time explicitly. We follow the same logic as before. Now consider Δy , the spatial interval in the y direction. As we consider a frame S' that moves in the positive x direction with speed u , this coordinate is not affected. The time component is always affected. We thus have:

$$v_y \equiv \frac{\Delta y}{\Delta t} = \frac{\Delta y'}{\gamma(\Delta t' + u\Delta x'/c^2)} = \frac{\Delta y'/\Delta t'}{\gamma(1 + u(\Delta x'/\Delta t')/c^2)} = \frac{v'_y}{\gamma(1 + uv'_x/c^2)}, \quad (32)$$

where I used $\Delta y'/\Delta t' \equiv v'_y$ and the same for x' as before. The same applied to the z direction, which results in the Einstein velocity transformations. For a frame S' moving with velocity u in the x direction we have

$$v_x = \frac{v'_x + u}{1 + uv'_x/c^2}, \quad (33)$$

$$v_y = \frac{v'_y}{\gamma(1 + uv'_x/c^2)}, \quad (34)$$

$$v_z = \frac{v'_z}{\gamma(1 + uv'_x/c^2)}. \quad (35)$$

Board Question

Frame S' moves to the right with $\beta = v/c = 3/5$ relative to frame S . In units with $c = 1$, an event P has coordinates

$$(t, x) = (10, 4)$$

in S .

1. Sketch the S and S' axes on a spacetime diagram and place the event P qualitatively.
2. Compute (t', x') using the Lorentz transformations.
3. Check that $t^2 - x^2 = t'^2 - x'^2$.
4. Explain in words why the diagram and the algebra agree about whether P is inside the future light cone of the origin.